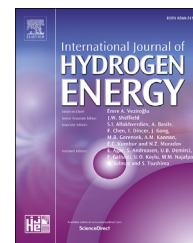


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# Development of large MH tank system for renewable energy storage

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## ABSTRACT

Metal Hydrides (MH) can absorb large quantities of hydrogen at room temperature and ordinary pressure. Because MH can store hydrogen at a pressure less than 0.1 MPa safely and compactly, it is looked to as a method of storing hydrogen produced by electricity derived from renewable energy sources. To study this method of storing renewable energy, we made a MH tank system which could store hydrogen in the range of 1000 Nm<sup>3</sup>. A Mm-Ni-Mn-Co alloy was used for this MH tank system. MH becomes pulverized with absorbing and desorbing hydrogen, and this causes the problem of MH tank transformation owing to the partial distribution of the pulverized MH powders. Our MH material, named “Hydrage™,” was made using a technique to compose the MH powders with polymer materials without decreasing the hydrogen absorption and desorption rate. With this technique, the MH powders were immobilized, and strain on the MH tank was reduced. Furthermore, this technique enabled uniform dispersion of the MH powders, and high-density filling in MH tank was achieved relative to that attainable in a conventional MH tank. An MH tank system with a capacity of 1000 Nm<sup>3</sup> is 1,800 mm in width, 3,150 mm in length, and 2,145 mm in height. The system for renewable energy storage consists of 9 tanks. About 7.2 tons of MH were used in this system. This system could work at temperatures from 25 to 35° C, and its maximum hydrogen absorption and desorption rate is 70 Nm<sup>3</sup>/h with a medium flow rate of 30 NL/min. This type of MH tank system, which can store a large amount of hydrogen safely and compactly, has the potential to become popular with various applications in the future.

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## Introduction

Renewable energy is becoming increasingly widespread due to the need for CO<sub>2</sub> emission and environmental load reductions. But if a power generation facility with large fluctuations were introduced widely, the electric power grid could become unstable. Therefore an electrical energy storage

technology is required. Hydrogen is thought to be one of the means for solving this problem. Hydrogen is suggested as the most suitable electrical energy storage medium in the field of large capacity and high output power devices [1]. Furthermore, hydrogen is useful for long-term electrical energy storage.

Metal Hydrides (MH) have attracted attention because they can store hydrogen most compactly among various methods

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of storing hydrogen. However, MH becomes pulverized with repeated cycles of absorbing and desorbing hydrogen, and this causes the problem of MH tank transformation owing to the partial distribution of the pulverized MH powders [2]. In order to prevent tank transformation, we made a composite where the alloy powder is mixed with resin. The tank was filled with the composite, and we evaluated the filling rate and the hydrogen absorption and desorption rate. Furthermore, we designed and produced a large MH tank using the composite.

## Experimental details

MmNi<sub>4.4</sub>Mn<sub>0.1</sub>Co<sub>0.5</sub> powders whose plateau is designed to be near atmospheric pressure under ambient temperature conditions were adopted. The initial powder size was about 1 mm and two types of resin were used for the composites. One was a hard resin (Type A), and the other was a soft resin (Type B). ELASTOSIL® M4648 was used for the hard resin, and WACKER SilGel® 612 was used for the soft resin. The MH and resin were mixed sufficiently to disperse the MH particles uniformly at 298 K. The weight ratio of the MH and the resin was 96.5:3.5. These composites were filled into a low-strength aluminum container 38 mm wide, 46.5 mm long and 10 mm high. The Type A composite was cut into a rectangular form and put into the aluminum container. The Type B composite was filled directly into the aluminum container because the composite became gelatinous. We investigated the microstructure, pressure-composition isotherms (P-C-T), and reaction rate of each composite. The measurements of hydrogen capacity, P-C-T characteristics, and reaction rate were carried out at 293 K and a hydrogen pressure of 0.001–5 MPa. Furthermore, the strain of various aluminum containers with hydrogen absorption and desorption was investigated.

## Results and discussion

Fig. 1 shows the microstructure of the composite. It was confirmed that the MH particles were dispersed uniformly in the composite. Fig. 2 shows two types of composite sheets. The one on the left is Type A and that on the right is Type B. The MH powder was dispersed uniformly and immobilized by mixing the MH and resin.

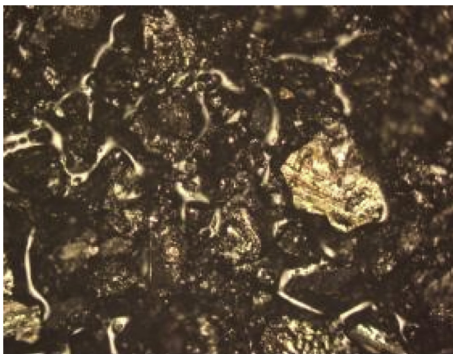


Fig. 1 – Microstructure of the composite.

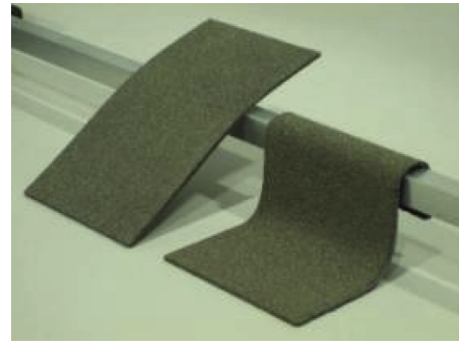


Fig. 2 – Appearance of two types of composites.

Fig. 3 shows the P-C-T curves of the alloy powder and the two types of composite at 293 K. No special change was observed in the PCT curves, and both the hydrogen equilibrium pressure and the hydrogen content were approximately equivalent. Watanabe et al. reported that the plateau pressure of MH fixed in resin was a little lower than that of the alloy powder alone and the hydrogen content of MH fixed in resin was limited to about 80% of that of alloy powder [3] alone. Watanabe et al. made the composites at temperatures of 393 K–453 K. Therefore, it was thought that the alloy surface was poisoned. But our composites were made at 298 K, so the alloy surface of our composites was not poisoned, and the plateau properties and hydrogen content of our composites were equivalent to that of alloy powder.

Fig. 4 shows the reaction rate of the activation at 293 K after degassing at 353 K for 2 h. The initial activation time of the composites was longer than that of the alloy powder because the active surface of the MH powder is decreased by mixing with resin. On the other hand, the reaction rates of the Type A and Type B composites were almost the same. From this result, the ratio of active surface of the two composites is supposed to be almost the same. Thus, the reaction rates of the composites at the initial activation were slower than that

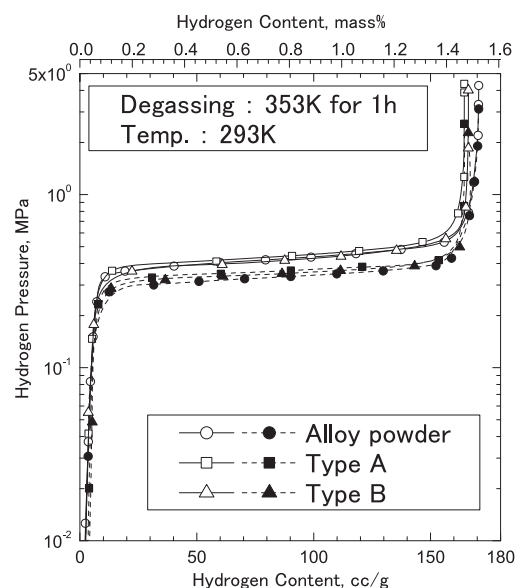


Fig. 3 – Difference of P-C-T characteristics.

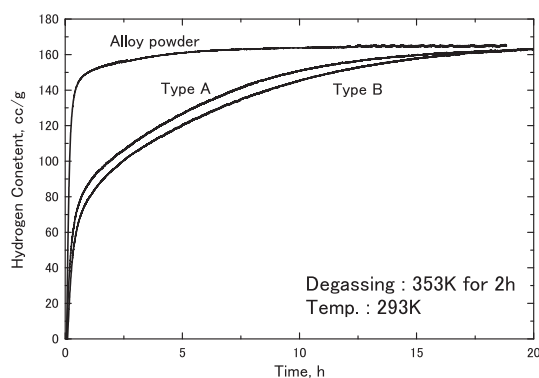


Fig. 4 – Difference of the first activation.

of alloy powder alone, but no difference was observed between the Type A and Type B composites.

Fig. 5 shows the second absorption and desorption rate of the Type B composite and alloy powder at 293 K. Each sample was poured into the aluminum container. Two conditions were selected for the amount of alloy powder used. In one condition, we used the same weight of powder as of the Type B composite. In the other condition, we used the same volume of powder as of the Type B composite. Compared with the alloy powder of the same weight, the hydrogen absorption rate of the Type B composite was slower than that of the alloy powder. But for desorption, considerable improvement was seen in the reaction rate of the Type B composite. These phenomena are thought to result from the newly formed surface appearing due to the pulverization caused by each cycle of hydrogen absorption and desorption. However, it was confirmed that the reaction rate of the Type B composite was slightly slower than that of the same weight of alloy powder, even in later cycles. On the other hand, compared to the alloy powder of the same volume, the Type B composite had a larger hydrogen content and a higher reaction rate than the alloy powder. These properties show that high-density filling was achieved using the composite. Thus, when the MH tank was filled with the Type B composite, it showed better characteristics than with the alloy powder.

Fig. 6 shows the shape retention properties of the composites. In both composites, no shape collapse was observed after 100 cycles of hydrogen absorption and desorption. It has

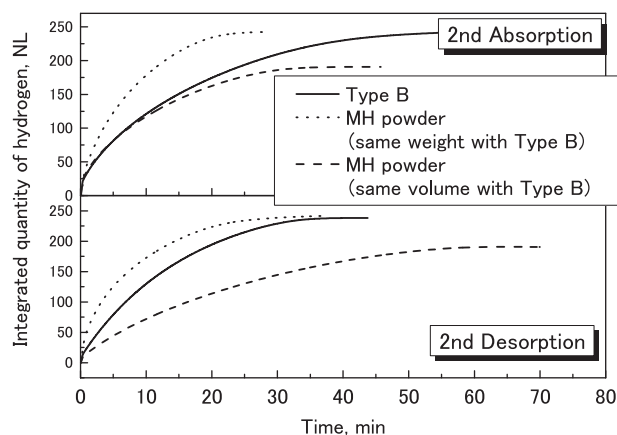


Fig. 5 – Difference of reaction rate.

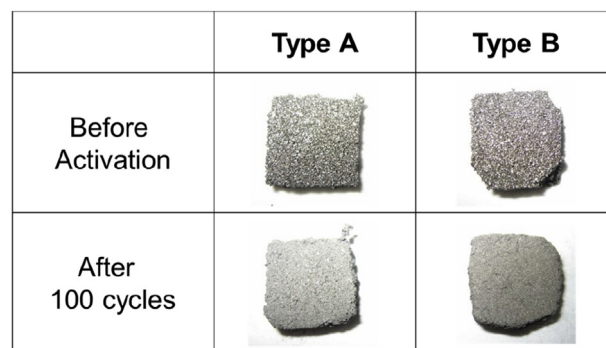


Fig. 6 – Shape retention property of composites.

been reported that the particle size of MH changes very little after tens of cycles [4] of absorbing and desorbing hydrogen, so the pulverization did not affect the shape of composites even in later cycles. Furthermore, cracks were not observed in the composites after the cyclic tests because the MH powder was immobilized in materials with high elasticity such as resin. This result suggests the composite has excellent cyclic characteristics and long-term durability.

Fig. 7 shows the relationship between the MH volume-filling ratio into the container and the strain on the aluminum container after 100 cycles. For the alloy powder, the maximum MH volume-filling ratio remained at 37%, and the strain on the container was very high. This level of strain caused plastic deformation in the container. On the other hand, for the composites, the maximum MH volume-filling ratio was from 50% to 55%, and the strain on the container was very low. The reason for the achievement of this high filling ratio when using the composites is thought to be that the MH powder was dispersed uniformly, and the MH powder in the composite did not move. In particular, the Type B (soft resin) composite had better properties. The higher filling ratio with the Type B composite relative to that of the Type A composite is thought to result from the fact that the hard resin was inserted into the container in rectangular form. Furthermore, the higher strain on the container wall of the Type A composite is thought to be due to the difference in the elastic coefficients of the resins. Thus, it was confirmed that the different resins show differences in filling ratio and strain. Nevertheless, both the transformation of the MH tank and the

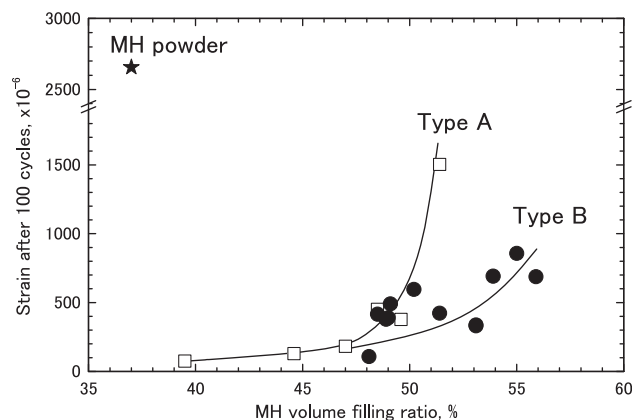


Fig. 7 – Relationship between filling ratio and strain.

filling ratio of the MH were improved by using the composites. These composites were named “Hydrage™” and are considered to be applicable to all MH tanks.

### Large MH tank

A large MH tank was designed based on a heat transfer calculation and some experiments. Eq. (1) is a basic heat exchange formula.  $Q$  is the heat capacity,  $U$  is the total heat transfer coefficient,  $A$  is the heat transfer area, and  $\Delta T$  is the temperature difference. Eq. (2) is Eq. (1) with both sides by divided by alloy weight,  $W$ . Eq. (2) was used to design the large MH tank.  $Q/W$  is the heat capacity per unit of MH alloy, and  $\Delta T$  is the temperature difference between the medium and the MH alloy.  $Q$ ,  $W$ , and  $\Delta T$  were obtained from the results of the hydrogen absorption and desorption test using the MH tank, so the  $UA/W$  value was obtained experimentally. The outer diameter of the tank and amount of MH alloy was decided by consideration  $UA/W$  values.

$$Q = UA\Delta T \quad (1)$$

$$Q_{H_2}/W = (UA/W)\Delta T \quad (2)$$

For the large MH tank, a double-tube type tank was designed using  $UA/W$  experimentally, and a little improvement was added. First, in order to increase the desorption rate, the MH alloy design was changed. Fig. 8 shows the P-C-T curves of two types of MH alloy. The alloy component was redesigned, the hydrogen equilibrium pressure of the MH was set higher, and a flat plateau was obtained. Second, in order to reduce pressure loss, the structure, arrangement, and area ratio of ventilation material were revised. As a result, pressure loss was greatly improved in the He-gas flow test. Fig. 9 shows the relationship between He-flow rate and pressure loss. With these improvements, the pressure loss decreased to one third. Finally, to enhance heat transfer, the interval of the heat transfer fins was shortened. Fig. 10 shows the improved MH

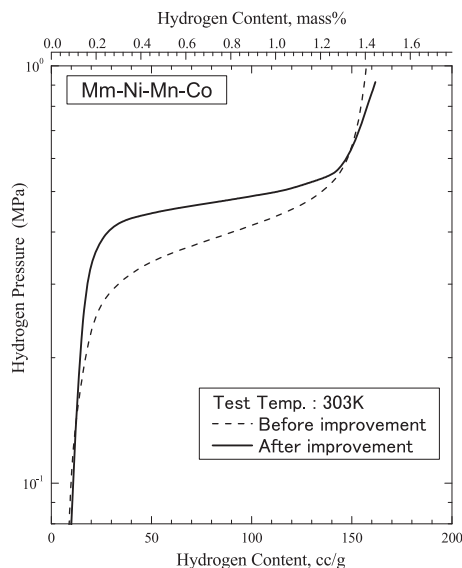


Fig. 8 – P-C-T characteristics of Metal Hydride for a large tank.

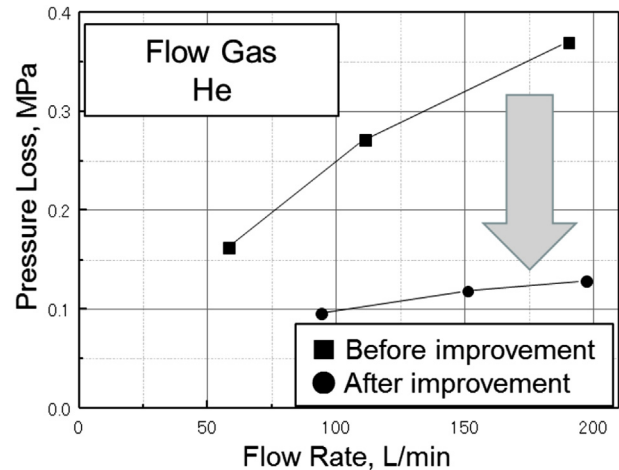


Fig. 9 – Relationship between gas flow rate and pressure loss of a large tank.

tank resulting from these design changes. A water-cooling jacket was welded on the outside of the MH tank, and ventilation and heat transfer fins were constructed. The tank was 242 mm in outer diameter and 1,780 mm in length. This tank was made of SUS304 stainless steel (medium side) and SUS316L (MH side), and obtained a certification of ASME CODE SECTION 8 DIVISION 1. One unit was comprised of two tanks, and each unit was 800 mm wide, 2,000 mm long, and 500 mm high. The hydrogen storage capacity of one unit was about 60 Nm<sup>3</sup>, and 19 units included 38 tanks were made for the hydrogen storage system. Thus, the total hydrogen storage capacity of this system exceeds 1100 Nm<sup>3</sup> equal to 100 kg of

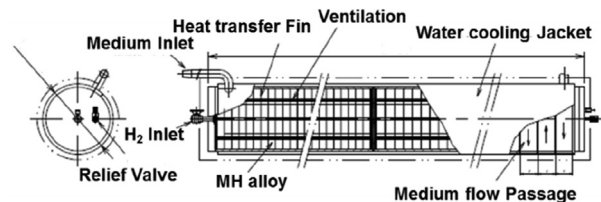


Fig. 10 – Drawing of a large MH tank.

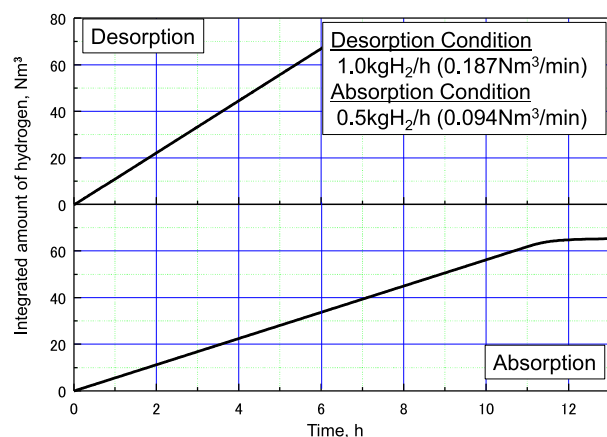


Fig. 11 – Hydrogen absorption and desorption rate of one unit.





**Fig. 12 – 1000 Nm<sup>3</sup>-class Hydrogen storage system.**

hydrogen. Fig. 11 shows the hydrogen absorption and desorption characteristics of one unit. The desorption rate was 0.187 Nm<sup>3</sup>/min (11.2 Nm<sup>3</sup>/h), and the absorption rate was 0.094 Nm<sup>3</sup>/min (5.6 Nm<sup>3</sup>/h). Almost 100% of the hydrogen content was absorbed and desorbed at the prescribed rate. No MH tank transformation was observed at all when using Hydrage™, and a high-volume-density hydrogen-storage system was enabled. This tank system was adopted as a hydrogen storage device of the system for renewable energy storage constructed at Griffith University [5].

Similar to the above, another 1,000 Nm<sup>3</sup>-class hydrogen storage system was considered. The outer diameter of this tank was increased from 242 mm to 340 mm and the tank was lengthened from 1,780 mm to 2,850 mm in order to allow slower hydrogen absorption and desorption rates than the above system. Thus, the hydrogen storage capacity of one tank increased compared with system discussed earlier. As a result, the overall number of tanks in this system could be reduced. Fig. 12 shows the 1000 Nm<sup>3</sup>-class hydrogen storage system. This system was 1,800 mm in width, 3,150 mm in length, and 2,145 mm in height. It consisted of 9 tanks and could store about 1000 Nm<sup>3</sup> of hydrogen. About 7.2 ton of MH

was used in this system. This system could work at temperatures from 25 to 35° C, with maximum hydrogen absorption and desorption rates of 70 Nm<sup>3</sup>/h at a medium flow rate of 30 NL/min. In this way, a compact tank system which could store a large quantity of hydrogen was developed.

## Conclusion

For high-density filling, a MH-resin composite was developed. The composite is named “Hydrage™” and registered as a trademark. MH tank transformation was suppressed using this technique and we were able to fill the tank with high-density MH. For renewable energy storage research, a large MH tank system with a double-tube structure was manufactured. The structure of the MH tank was improved, including the MH alloy composition, ventilation, and heat transfer fins. We continue to develop MH tanks that can be adapted to various demands, and hope that this MH tank system will be used in various applications and will become popular in the future.

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